

实验内容

- 测试 PRIMA：一种无导数的非线性优化求解器
- Rosenbrock 函数在 4 种约束条件下验证正确性
- 474 个 S2MPJ 问题，对比 single / double / quadruple 精度

工具链：PRIMA + OptiProfiler + S2MPJ, MATLAB R2025b, Linux

遇到的困难

S2MPJ 缺失

- 报错 `No problem is selected`, 排查发现未安装问题库

MATLAB 内存耗尽崩溃

- 运行中途被系统 OOM-kill, Matlab 被系统强制终止运行。

运行失败文件

-  matlab_crash_dump.5421-1
-  matlab_crash_dump.9873-1
-  matlab_crash_dump.14732-1
-  matlab_crash_dump.49392-1
-  matlab_crash_dump.52500-1
-  matlab_crash_dump.53440-1
-  matlab_crash_dump.55673-1
-  matlab_crash_dump.62890-1
-  matlab_crash_dump.69356-1
-  matlab_crash_dump.81054-1

运行结果文件夹

prima_double_prima_quadruple_u0ln_2_10_0_10_noisy_0.001_mixed_gaussian_20200505_182443

prima_double_prima_quadruple_u0ln_2_10_0_10_noisy_0.001_mixed_gaussian_20200505_221603

prima_double_prima_quadruple_u0ln_2_10_0_10_plain_20200505_232818

prima_double_prima_single_u0ln_2_10_0_10_noisy_0.001_mixed_gaussian_20200504_104633

prima_double_prima_single_u0ln_2_10_0_10_noisy_0.001_mixed_gaussian_20200506_010035

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_060341

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_065401

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_101541

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_145141

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_145534

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_150442

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_150932

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_151656

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_152536

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_164704

prima_double_prima_single_u0ln_2_10_0_10_plain_20200504_123558

prima_double_prima_single_u0ln_2_10_0_10_plain_20200504_183729

prima_double_prima_single_u0ln_2_10_0_10_plain_20200504_230947

prima_double_prima_single_u0ln_2_10_0_10_plain_20200504_231016

prima_double_prima_single_u0ln_2_10_0_10_plain_20200504_231105

prima_double_prima_single_u0ln_2_10_0_10_plain_20200505_090832

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_191956

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_195403

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_195641

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_200036

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_214739

prima_double_prima_single_u0ln_2_20_0_10_noisy_0.001_mixed_gaussian_20200503_214803

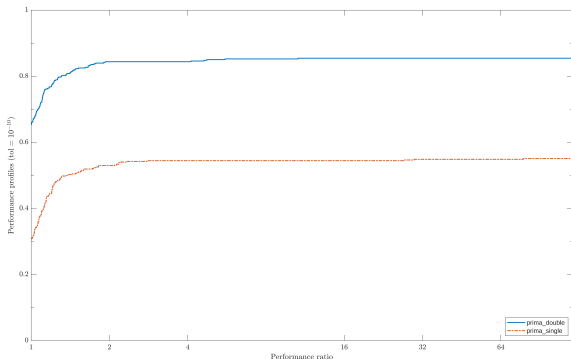
解决方案

- AI 辅助优化 `exporthist.m`。
- 扩大虚拟机内存与 swap 空间
- 串行改为 8 核并行池 (`parpool`)
- 将 benchmark 的 `maxdim` 从 20 降至 10
- 关闭图形化界面改为纯命令行运行。

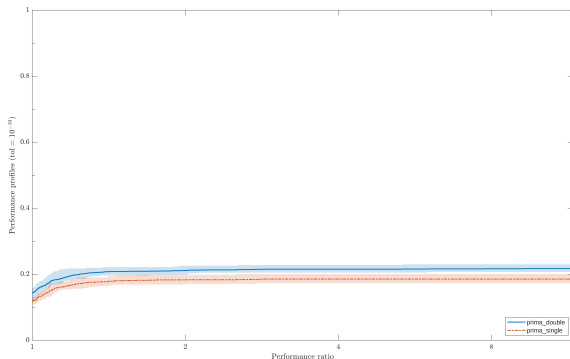
实验结果

Rosenbrock ——四种约束条件下均收敛至合理最优值

S2MPJ 精度对比——double 与 single 存在可观测差距，double 与 quadruple 性能曲线基本重合



plain



noisy